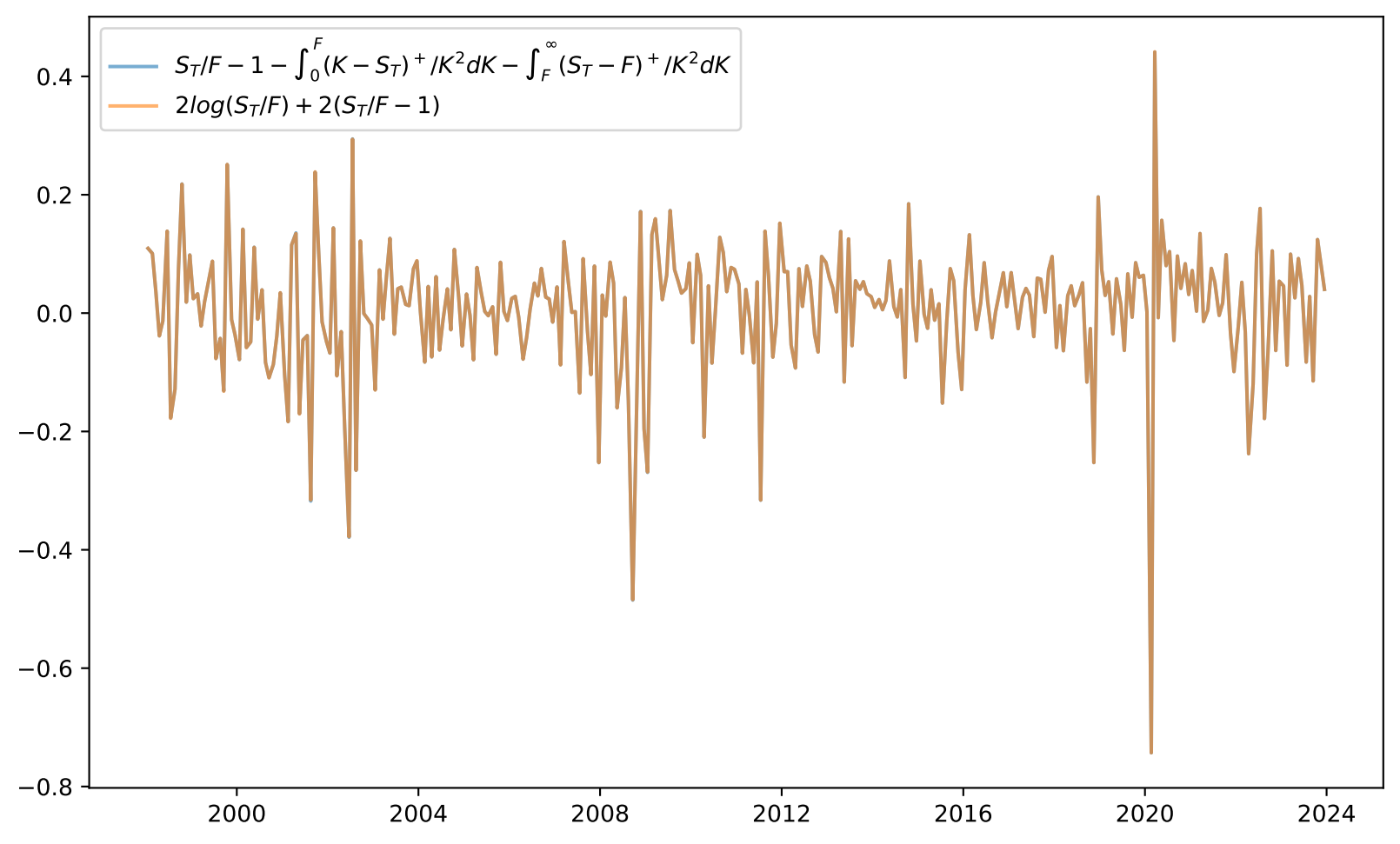
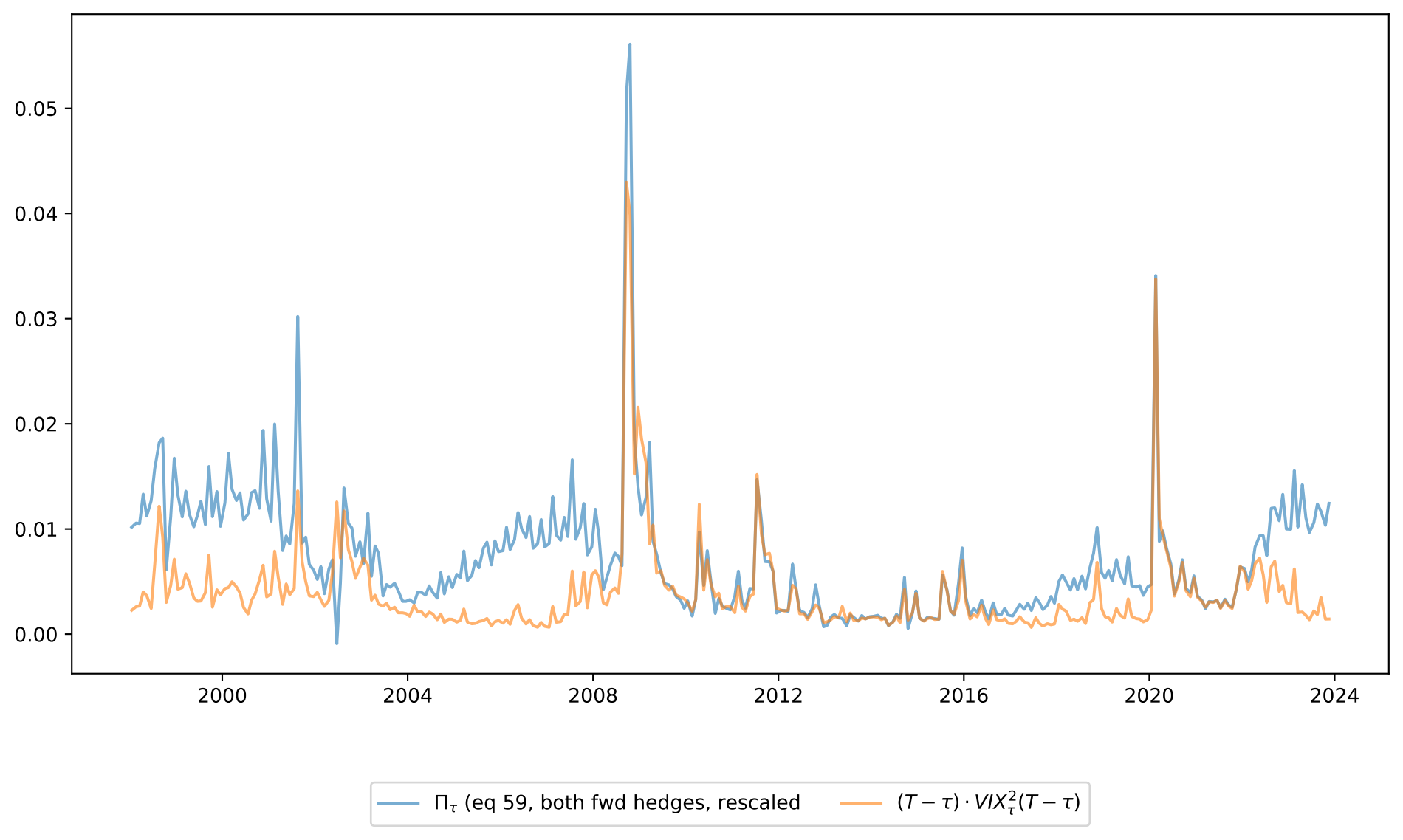

$S_T/F - 1 - \int_0^F (K - S_T)^+ / K^2 dK - \int_F^\infty (S_T - F)^+ / K^2 dK$
$2log(S_T/F) + 2(S_T/F - 1)$
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
2000
2004
2008
2012
2016
2020
2024

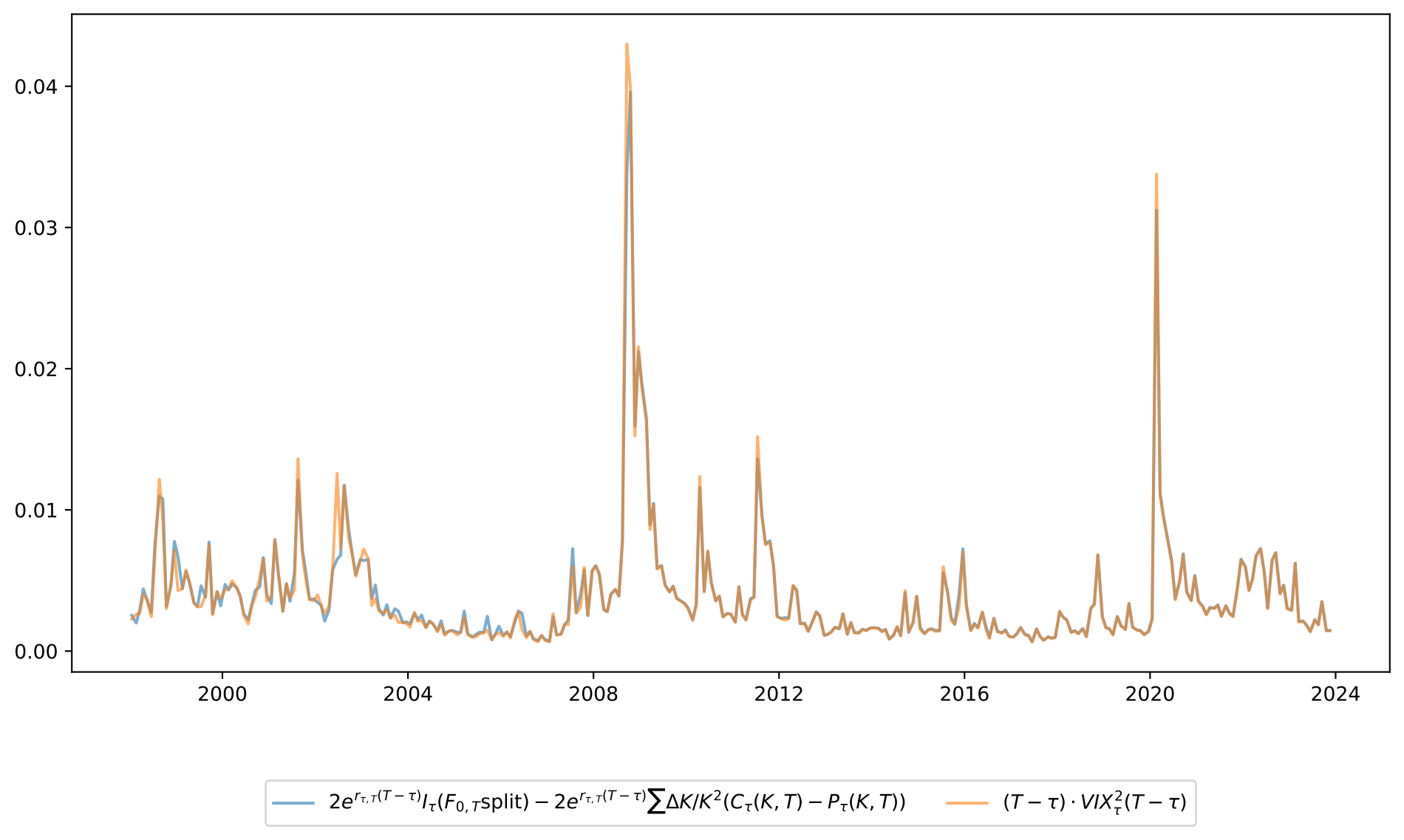
log_actual	log_approx
1.0	0.99999820414581 5
0.99999820414581 5	1.0

log_actual	log_approx
312.0	312.0
0.007342	0.007341
0.109486	0.109533
-0.742954	-0.743079
-0.038563	-0.038554
0.024068	0.024157
0.069941	0.070041
0.441647	0.441112


0.05
0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
Π_τ (eq 59, both fwd hedges, rescaled
$(T-\tau)\cdot VIX_\tau^2(T-\tau)$

approximation_full_hedge	future_VIX^2
1.0	0.763843583278373 3
0.763843583278373 3	1.0

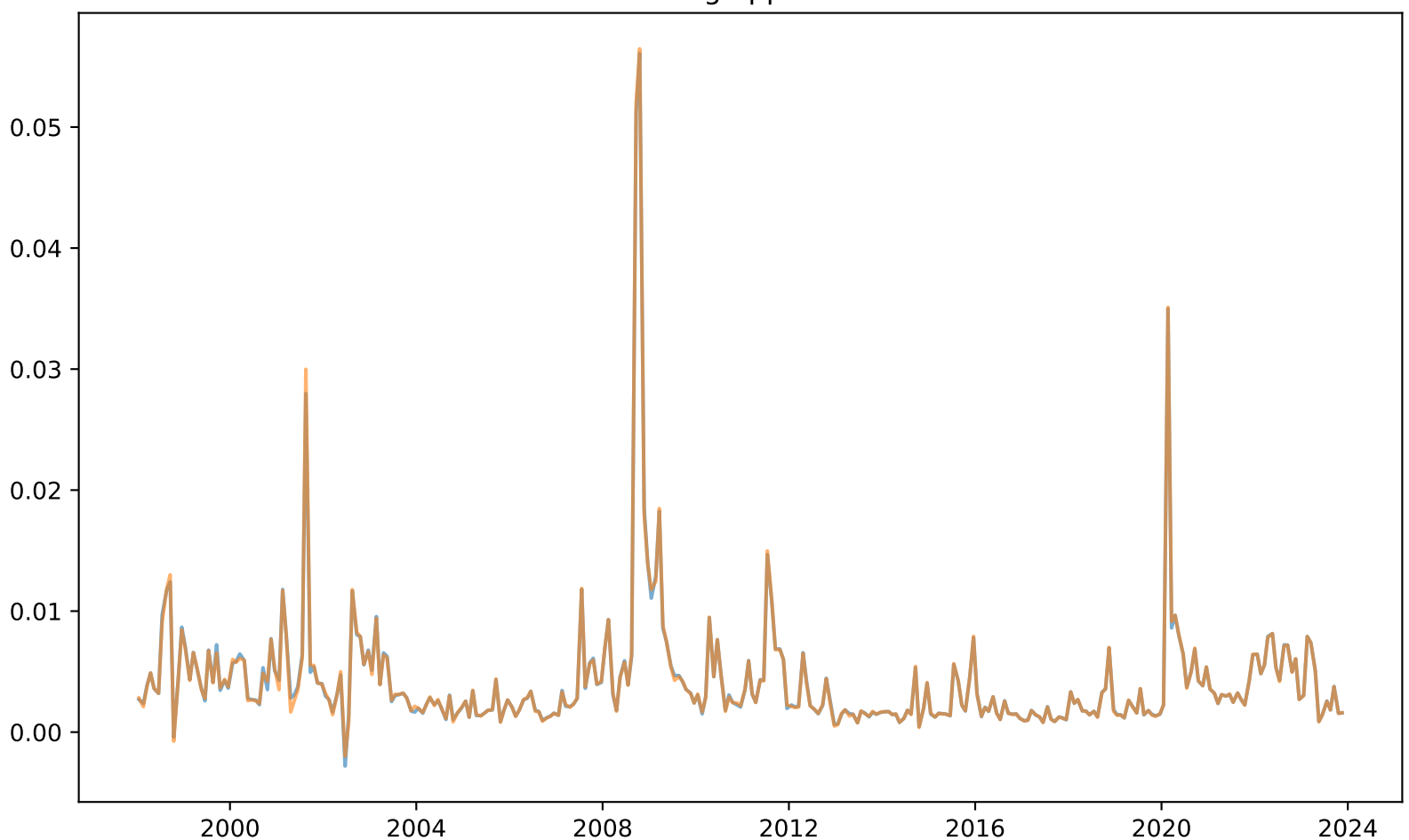
approximation_full_hedge	future_VIX^2
311.0	311.0
0.007134	0.0039
0.006037	0.004552
-0.000911	0.000638
0.003191	0.001549
0.005685	0.002676
0.010068	0.004441
0.056098	0.042988

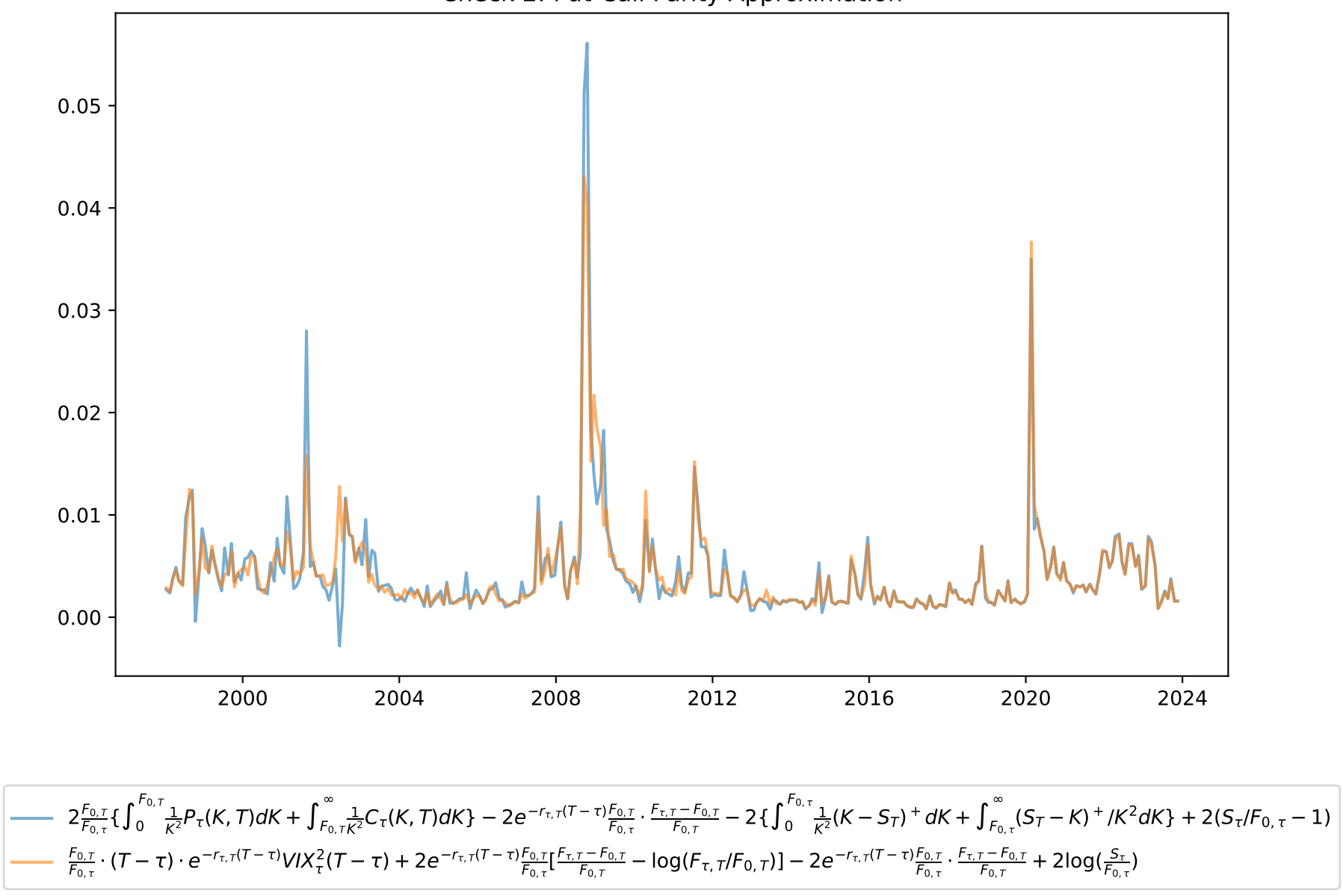

0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
$2e^{r_{\tau,T}(T-\tau)}I_\tau(F_{0,T}\text{split}) - 2e^{r_{\tau,T}(T-\tau)}\sum \Delta K/K^2(C_\tau(K,T) - P_\tau(K,T))$
$(T-\tau)\cdot VIX^2_\tau(T-\tau)$

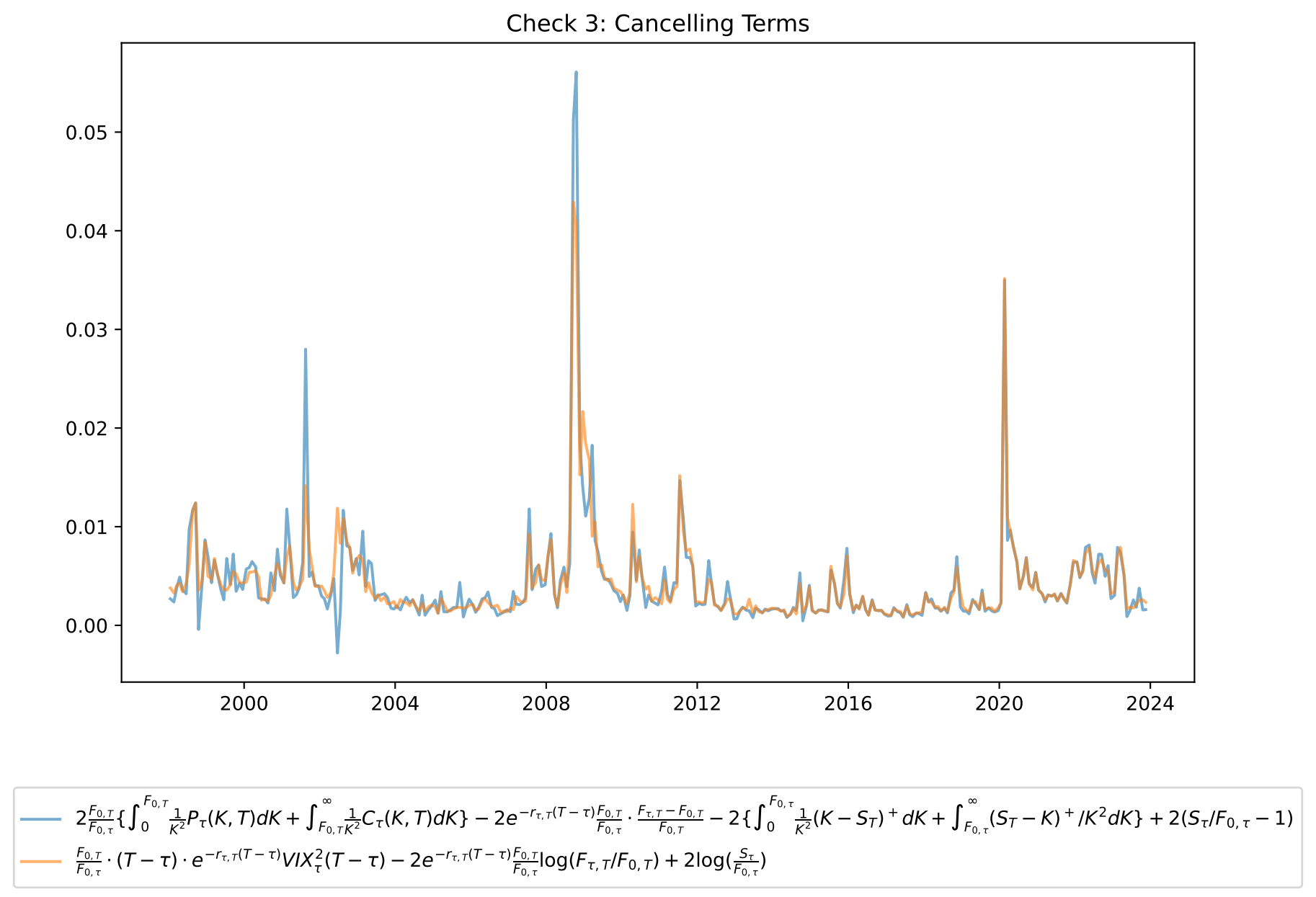
approximation_substitution	future_VIX^2
1.0	0.989346239631706 2
0.989346239631706 2	1.0

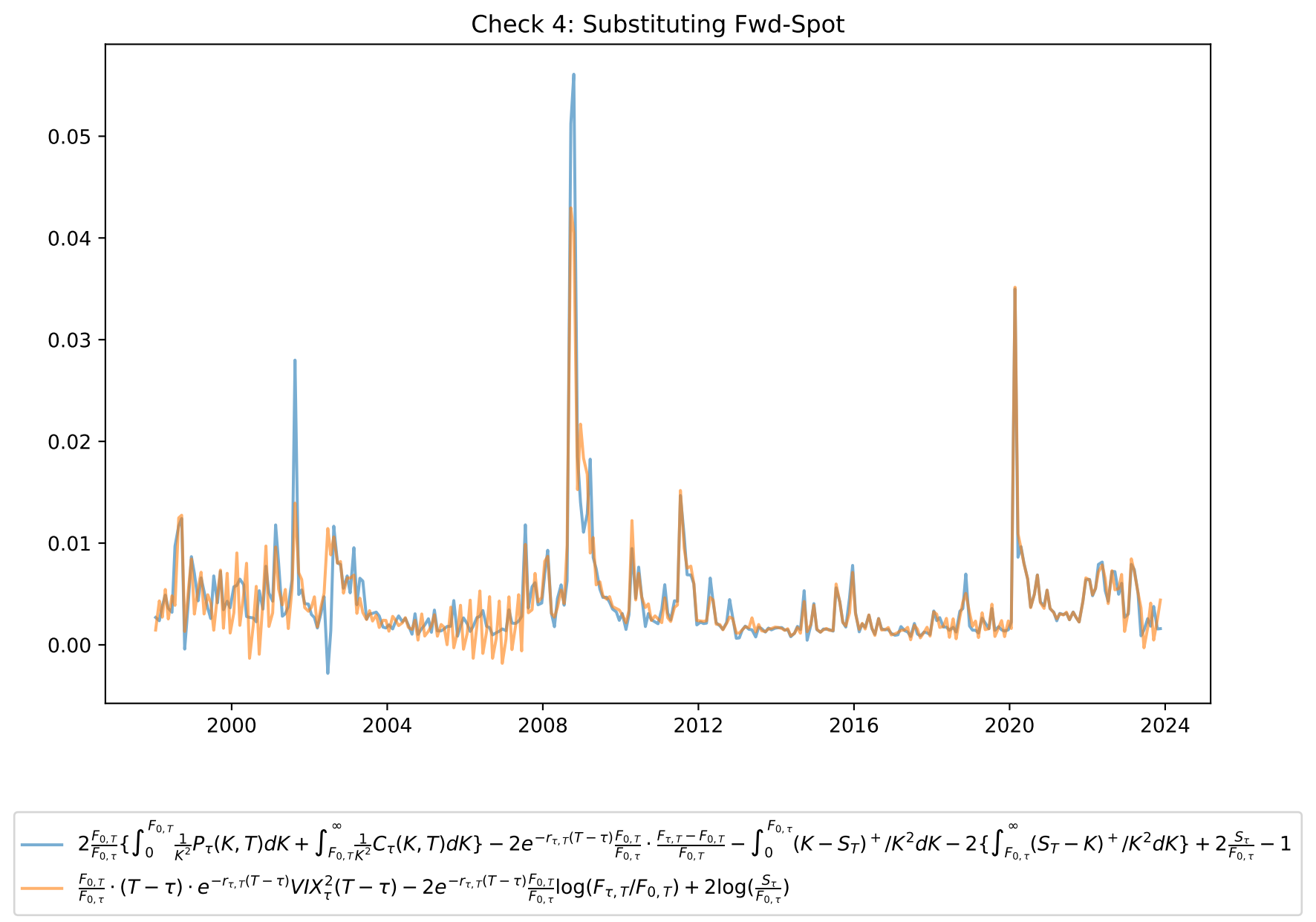
approximation_substitution	future_VIX^2
311.0	311.0
0.003907	0.0039
0.004217	0.004552
0.00066	0.000638
0.001608	0.001549
0.002757	0.002676
0.004609	0.004441
0.039583	0.042988

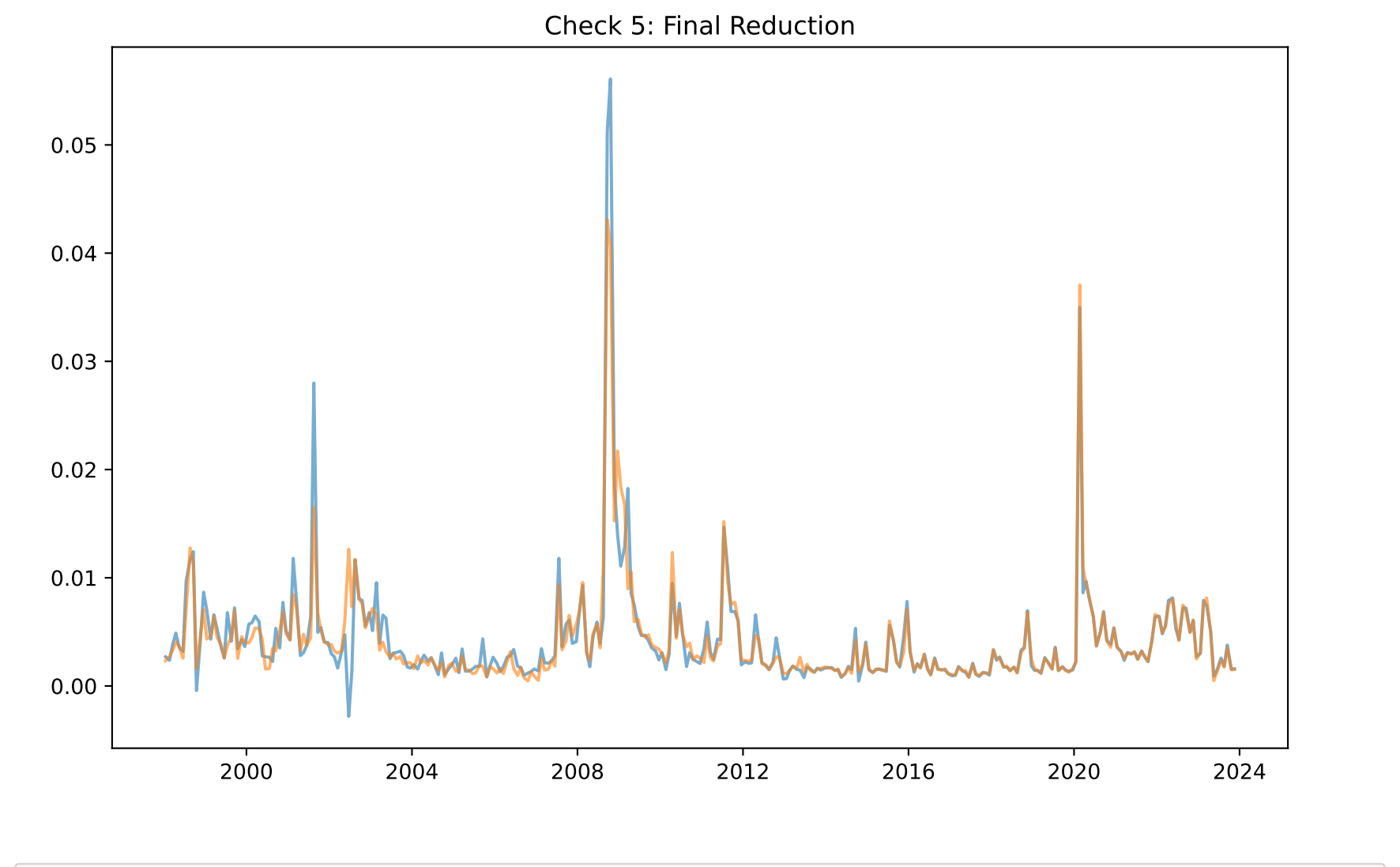
option_substitution_coverage
296.0
0.9999
0.0014
0.9762
1.0
1.0
1.0
1.0

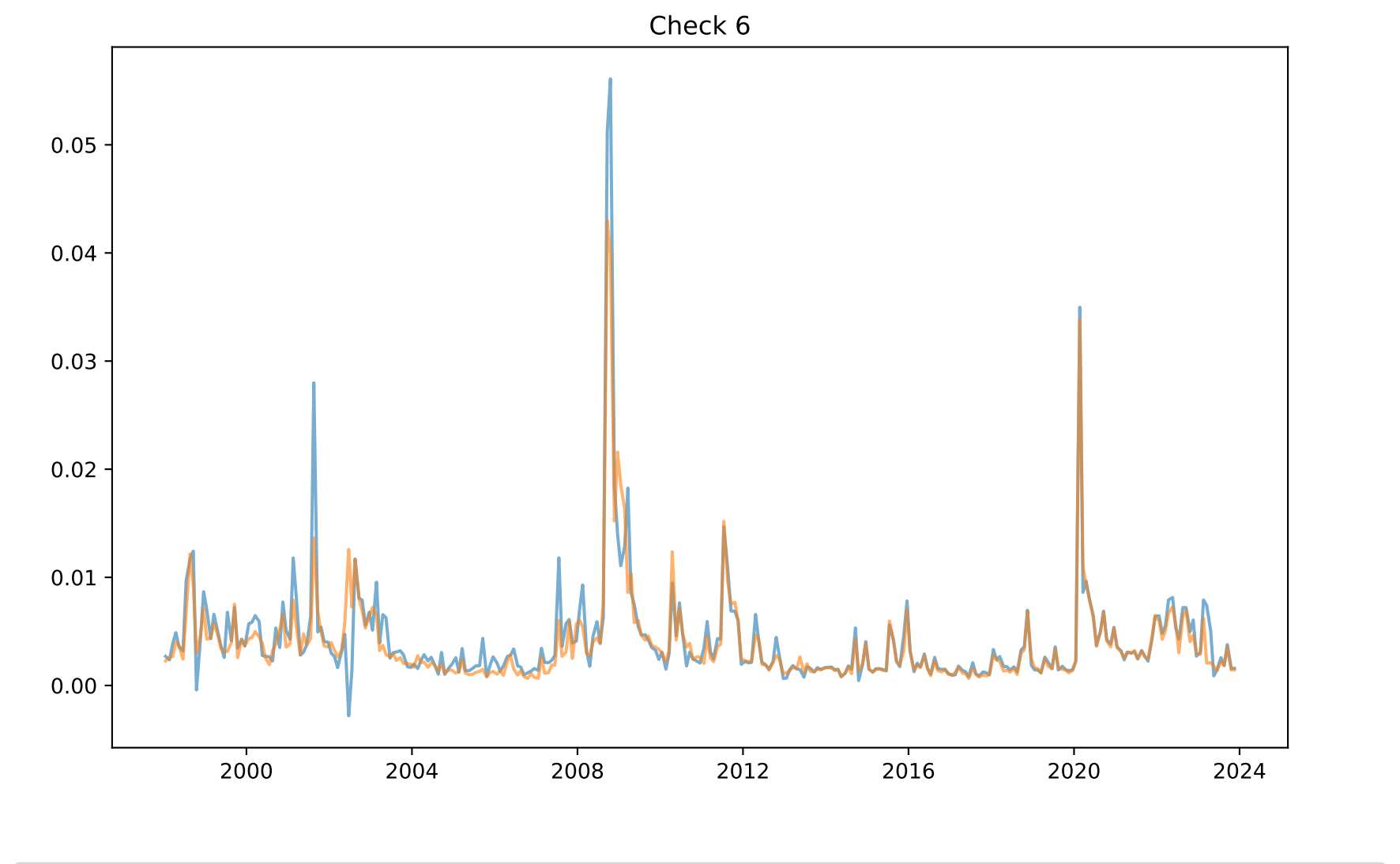

Check 1: log approximation
0.05
0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-2\{\int_0^{F_{0,\tau}}\frac{1}{K^2}(K-S_T)^+dK+\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2(S_\tau/F_{0,\tau}-1)$
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}+2\log(S_\tau/F_{0,\tau})$


Check 2: Put-Call Parity Approximation
0.05
0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-2\{\int_0^{F_{0,\tau}}\frac{1}{K^2}(K-S_T)^+dK+\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2(S_\tau/F_{0,\tau}-1)$
$\frac{F_{0,T}}{F_{0,\tau}}\cdot(T-\tau)\cdot e^{-r_{\tau,T}(T-\tau)}VIX_\tau^2(T-\tau)+2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}[\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-\log(F_{\tau,T}/F_{0,T})]-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}+2\log(\frac{S_\tau}{F_{0,\tau}})$


Check 3: Cancelling Terms
0.05
0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-2\{\int_0^{F_{0,\tau}}\frac{1}{K^2}(K-S_T)^+dK+\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2(S_\tau/F_{0,\tau}-1)$
$\frac{F_{0,T}}{F_{0,\tau}}\cdot(T-\tau)\cdot e^{-r_{\tau,T}(T-\tau)}VIX_\tau^2(T-\tau)-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\log(F_{\tau,T}/F_{0,T})+2\log(\frac{S_\tau}{F_{0,\tau}})$


Check 4: Substituting Fwd-Spot
0.05
0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-\int_0^{F_{0,\tau}}(K-S_T)^+/K^2dK-2\{\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2\frac{S_\tau}{F_{0,\tau}}-1$
$\frac{F_{0,T}}{F_{0,\tau}}\cdot(T-\tau)\cdot e^{-r_{\tau,T}(T-\tau)}VIX_\tau^2(T-\tau)-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\log(F_{\tau,T}/F_{0,T})+2\log(\frac{S_\tau}{F_{0,\tau}})$


Check 5: Final Reduction
0.05
0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-\int_0^{F_{0,\tau}}(K-S_T)^+/K^2dK-2\{\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2\frac{S_\tau}{F_{0,\tau}}-1$
$\frac{F_{0,T}}{F_{0,\tau}}\cdot(T-\tau)\cdot e^{-r_{\tau,T}(T-\tau)}VIX_\tau^2(T-\tau)-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\log(F_{\tau,T}/F_{0,T})+2\log(\frac{S_\tau}{F_{0,\tau}})$


Check 6
0.05
0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-\int_0^{F_{0,\tau}}(K-S_T)^+/K^2dK-2\{\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2\frac{S_\tau}{F_{0,\tau}}-1$
$(T-\tau)\cdot VIX_\tau^2(T-\tau)$

raw_portf_value	future_VIX^2
1.0	0.926375461923062
0.926375461923062	1.0

	raw_portf_value	future_VIX^2
count	311.0	311.0
mean	0.004251	0.0039
std	0.005413	0.004552
min	-0.002806	0.000638
25%	0.0017	0.001549
50%	0.002921	0.002676
75%	0.005084	0.004441
max	0.056099	0.042988

	% error
count	311.0
mean	13.264886
std	37.917218
min	-122.309934
25%	-3.692639
50%	6.478213
75%	23.05723
max	259.814062